
Do Post-Training Algorithms Actually Differ on Mathematical Reasoning at Scale?

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Abstract

1 Practitioners routinely benchmark post-training algorithms at small scale, then
2 deploy the winner at large scale. We frame the question as a *cross-scale study of*
3 *post-training algorithms*—tracking how algorithm rankings evolve as a function
4 of model size, without fitting a parametric scaling law—and show small-scale
5 benchmarking is *anti-predictive* for mathematical reasoning: the within-family
6 ordering of methods inverts as models grow. In a controlled study spanning
7 ~ 350 training runs across two model families (Qwen 2.5 0.5B–14B, Gemma 3
8 1B–12B), 8 algorithms, and 20 DPO variants with up to $N=5$ seeds, we find
9 three results. **(1) Rankings invert with scale:** at ≤ 3 B, Self-Play RFT (SP-RFT;
10 standard rejection sampling fine-tuning (Yuan et al., 2023) on the model’s own
11 correct trajectories) is the strongest offline method (at 3B under per-algorithm
12 best learning rates; Appendix O); at 7B+, SP-RFT collapses to worst while DPO
13 leads ($p < 0.001$). The inversion replicates in Gemma 3 and persists at Qwen2.5-
14 14B. **(2) Initialization dominates algorithm choice:** switching Instruct→Base
15 compresses the spread 3–15 $\times$. **(3) Apparent differences have two regimes:**
16 format compliance at small scale, architecture-specific reasoning at large scale.
17 Together, these findings imply that small-scale algorithm selection is unreliable for
18 mathematical reasoning, and practitioners should validate at deployment scale.

19 1 Introduction

20 Suppose you are choosing a post-training algorithm for mathematical reasoning. You benchmark
21 Self-Play RFT (SP-RFT; rejection sampling fine-tuning (Yuan et al., 2023) on self-generated correct
22 trajectories), Direct Preference Optimization (DPO (Rafailov et al., 2023)), and Simple Preference
23 Optimization (SimPO (Meng et al., 2024)) at 1.5B parameters: SP-RFT wins by 5 pp over DPO. You
24 select SP-RFT, scale to 7B, and deploy—only to discover that SP-RFT is now the *worst* method by
25 6 pp, while DPO has silently become best. Your small-scale selection was not just uninformative;
26 for the SFT-vs-preference axis it was anti-predictive—small-scale ranks invert at deployment scale
27 (Figure 1).

28 **We frame this as a cross-scale study of post-training algorithms:** rather than chase a single
29 best loss function, we measure how algorithm rankings evolve with model size and treat the curves
30 themselves—not the point estimates—as the object of study. We use the term *scaling study* (not
31 scaling *law*) deliberately: we do not fit a parametric form for accuracy(N), since the within-Qwen
32 DPO–SP-RFT gap narrows from +6.0 pp at 7B to +2.2 pp at 14B—a non-monotone behavior that
33 any single power/log fit would obscure.

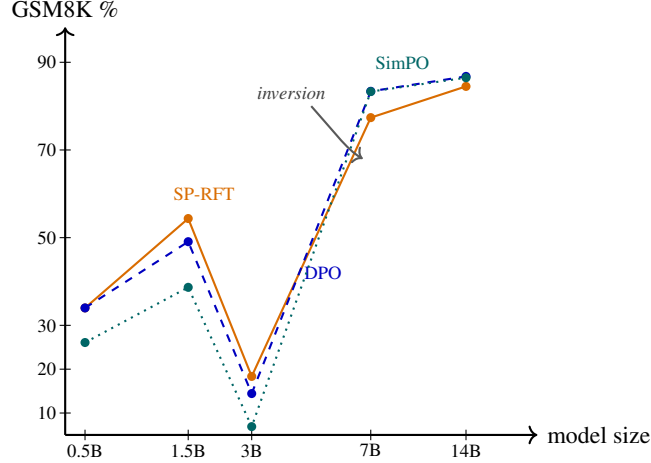


Figure 1: **Cross-scale ranking of post-training algorithms.** GSM8K strict-match accuracy vs. model size (log axis) for three offline post-training methods on Qwen 2.5, all at the shared default LR 10^{-6} . The deep dip at 3B is a checkpoint-specific format-compliance pathology of Qwen 2.5-3B-Instruct (Appendix O); the substantive crossover is between 1.5B (SP-RFT leads) and 7B+ (DPO/SimPO lead). Small-scale rankings are not predictive of deployment-scale rankings; the inversion replicates on Gemma 3 (1B–12B; Section 4.2).

34 This is not a hypothetical. It is the central empirical finding of this paper, replicated across two model
 35 families and confirmed with formal statistical testing ($p < 0.001$, $N=5$ seeds). The practitioner’s
 36 dilemma is real: every published comparison operates at a single scale, yet deployment happens at a
 37 different one. If rankings invert, the entire methodology of algorithm benchmarking at small scale is
 38 called into question.

39 Post-training of large language models—adapting a pretrained model for tasks such as mathematical
 40 reasoning—has become the decisive stage that converts a pretrained checkpoint’s raw problem-solving
 41 ability into the reliably formatted, instruction-following behavior that evaluation and deployment
 42 require (Ouyang et al., 2022; Touvron et al., 2023). Many algorithms compete for adoption: online
 43 RL methods like Group Relative Policy Optimization (GRPO (Shao et al., 2024)), offline preference
 44 methods like DPO, and a growing family of variants—SimPO, Kahneman-Tversky Optimization
 45 (KTO (Ethayarajh et al., 2024)), Odds-Ratio Preference Optimization (ORPO (Hong et al., 2024))—
 46 each claiming improvements over prior work. DeepSeek-R1 (DeepSeek-AI, 2025) demonstrated
 47 that pure RL post-training can elicit sophisticated reasoning, further expanding the design space. Yet
 48 each paper evaluates on different models, datasets, and scales, making cross-method comparison
 49 unreliable. The few existing studies (Xu et al., 2024; Spangher et al., 2025) cover limited methods at
 50 a single scale—they cannot detect the ranking inversions we observe.

51 We address this gap through a large-scale, controlled comparison using a unified framework where
 52 all algorithms share identical infrastructure. Our thesis is sharp: **on mathematical reasoning, post-
 53 training algorithm rankings invert with scale, making small-scale selection anti-predictive.** We
 54 emphasize that this is a task-specific phenomenon: on multiple-choice general benchmarks (ARC-C,
 55 HellaSwag, WinoGrande), algorithms differ by <1.2 pp at all scales (Section 4.4; Appendix T),
 56 though generative benchmarks remain untested. The evidence for mathematical reasoning unfolds in
 57 three findings:

- 58 1. **Rankings invert with scale—from Instruct checkpoints, across architectures.** On Qwen 2.5
 59 Instruct, SP-RFT leads offline methods at $\leq 3B$ but collapses to worst at 7B, where DPO
 60 ($83.4\% \pm 0.56$) and SimPO ($83.3\% \pm 1.79$) dominate ($p < 0.001$ vs. SP-RFT, $N=5$). On
 61 Gemma 3 IT, the same pattern holds: SP-RFT $>$ DPO at 1B and 4B, but DPO $\geq$ SP-RFT at 12B
 62 (87.4% vs. 86.7%).

- 63 2. **Initialization explains most of the variance.** Replacing Instruct with Base initialization collapses
 64 the inter-algorithm spread from 15.7 pp to 1.07 pp at 1.5B and from 6.00 pp to 1.82 pp at 7B (a
 65 $3\text{--}15\times$ compression). The dominant source of inter-algorithm variance is how each loss function
 66 interacts with prior alignment, not intrinsic properties of the objectives.
- 67 3. **Apparent differences have two regimes: format compliance at small scale, architecture-**
 68 **specific reasoning at large scale.** At 1.5B, the 19.3 pp GSM8K spread collapses $36\times$ to 0.54 pp on
 69 MATH and $41\times$ to 0.47 pp on general benchmarks—algorithms differ only in format compliance.
 70 At larger scales ($\geq 4\text{B}$), MATH reveals genuine reasoning differences, but these depend on the
 71 (algorithm, architecture) pair: DPO gains +9 pp on Gemma 12B MATH while SimPO gains
 72 +13 pp on Qwen2.5-14B MATH. Both regimes yield the same practical conclusion: small-scale
 73 benchmarking is unreliable.

74 These findings are based on ~ 323 training runs on Qwen 2.5 (0.5B–14B), supplemented by 27 runs
 75 on Gemma 3 (1B–12B), spanning 8 algorithms, 20 DPO variants, 4 evaluation domains (GSM8K,
 76 MATH, ARC-C/HellaSwag/WinoGrande), and up to $N=5$ seeds with formal statistical testing.
 77 The implication for practitioners is direct: validate at deployment scale, not prototype scale. The
 78 implication for the research community is that designing new loss functions at small scale—the
 79 dominant paradigm—may be selecting for the wrong properties.

80 2 Related Work

81 **Post-training algorithms and comparative studies.** The RLHF pipeline (Ouyang et al., 2022)
 82 uses PPO (Schulman et al., 2017); GRPO (Shao et al., 2024) simplifies this via group-normalized
 83 rewards. Wu et al. (2025) show a theoretical correspondence between GRPO and DPO; our empirical
 84 results show this correspondence is scale-conditioned (SGRPO leads DPO by +8.9 pp at 1.5B but
 85 trails by -2.8 pp at 7B), illustrating that scale-dependent empirical separations can coexist with their
 86 scale-free theoretical equivalence. DPO (Rafailov et al., 2023) bypasses reward modeling, spawning
 87 IPO (Azar et al., 2024), SimPO (Meng et al., 2024), KTO (Ethayarajh et al., 2024), ORPO (Hong
 88 et al., 2024), and others. We provide the first controlled evaluation of 20 DPO variants under identical
 89 conditions. Xu et al. (2024) and Ivison et al. (2024) cover only 2–3 algorithms. Spangher et al.
 90 (2025) benchmark 17 algorithms at a single scale—they cannot detect ranking inversions. Saeidi et al.
 91 (2025) evaluate DPO variants without multi-seed runs or multiple comparison correction. Our finding
 92 that variant rankings are scale-dependent is reminiscent of Lucic et al. (2018), who found that GAN
 93 variant rankings change across experimental conditions. Critically, none of these studies evaluate
 94 across multiple model families, leaving the question of architectural generality unanswered. Our
 95 Gemma 3 validation addresses this gap. Tan et al. (2025) study scaling behaviors of RL post-training;
 96 our work complements theirs by showing that *relative* rankings—not just absolute performance—are
 97 scale-dependent.

98 3 Methods and Experimental Setup

99 **Framework.** Our framework implements 51 post-training algorithms (the full list is enumerated
 100 in [anon repo]; this paper evaluates 28: 8 core methods plus 20 DPO variants, with the remaining
 101 23 archived for future comparisons). All algorithms share identical model loading (HuggingFace,
 102 `flash_attention_2`), data pipelines, distributed training (DeepSpeed ZeRO-3 (Rajbhandari et al.,
 103 2020), vLLM (Kwon et al., 2023) for RL rollouts), and checkpointing. Each algorithm implements a
 104 `BaseAlgorithm` interface requiring only the loss function, ensuring the *only* difference between runs
 105 is the loss computation. We organize algorithms into: (1) **SP-RFT**; (2) **Online RL** (GRPO variants);
 106 (3) **Offline preference** (DPO (Rafailov et al., 2023), IPO (Azar et al., 2024), SimPO (Meng et al.,
 107 2024), KTO (Ethayarajh et al., 2024), and 16 additional variants); (4) **Hybrid** (ORPO (Hong et al.,
 108 2024), SPIN (Chen et al., 2024)). The 20-variant taxonomy is in Appendix A. **SGRPO** (token-level
 109 GRPO with on-policy rollouts, exact-match reward, group-normalized advantages, $\lambda_{\text{KL}}=0$; adding

110 $\lambda_{KL}=0.01$ yields +0.15 pp per Appendix P) is the online-RL representative in all main tables. **GSPO**
 111 (sequence-level importance ratio) and **CISPO** (detached-weight clip) are ablations in Appendix E.

112 **Evaluation.** GSM8K (Cobbe et al., 2021) (1,319 test problems, exact-match, greedy decoding).
 113 MATH (Hendrycks et al., 2021) (5,000 problems, 4-shot) at 1.5B and general-domain benchmarks
 114 (ARC-C, HellaSwag, WinoGrande) at both 1.5B and 7B. All evaluations use the LM Evaluation
 115 Harness (Gao et al., 2024). We report means $\pm\sigma$ and perform Welch’s t -tests with Bonferroni
 116 correction where indicated.

117 **Qwen 2.5 (primary).** Qwen 2.5 (Qwen Team, 2025) Instruct at 0.5B, 1.5B, 3B, 7B, and 14B. Our
 118 experiments comprise: (1) core comparison of 8 algorithms across 4 scales (~ 70 runs, $N=5$ seeds
 119 for core methods at 7B), with a full-FT/LoRA factorial at both 3B and 7B; (2) DPO variant taxonomy
 120 at 1.5B (20 variants $\times$ 5 seeds = 100 runs) and a stratified subset of 5 at 7B (15 runs); (3) 14B
 121 deployment-scale validation with SP-RFT, DPO, SimPO ($N=3$ seeds each); (4) ablations (~ 57 runs),
 122 including Base-vs-Instruct initialization and hyperparameter sensitivity; (5) a fixed-dataset control
 123 training 1.5B on 14B self-play data (6 runs, Section 4.5); (6) LR sweeps at 1.5B, 3B, and 7B (54
 124 runs, Appendix M–O); (7) a gold-data SFT control using MetaMathQA at 1.5B and 7B (6 runs,
 125 Section 4.5). Total Qwen-side: ~ 323 training runs, ~ 640 GPU-hours on H100s (training only; the
 126 ~ 940 GPU-hour grand total in Appendix I also includes evaluation). Together with the Gemma 3
 127 runs below, the grand total is ~ 350 training runs.

128 **Gemma 3 (cross-architecture validation).** To test whether our findings generalize beyond Qwen,
 129 we train SP-RFT, DPO, and SimPO on Gemma 3 (Gemma Team, 2025) IT at 1B, 4B, and 12B with
 130 3 seeds each (27 runs). This uses the same GSM8K data and evaluation protocol; 12B uses LoRA,
 131 matching the Qwen2.5-7B setup.

132 **Data.** All algorithms share the same data pipeline to eliminate data-source confounds. For each
 133 prompt in GSM8K’s 7,473 training problems, we sample 16 responses from the *target model* at
 134 temperature 1.0 and score each with exact-match correctness. DPO/SimPO/IPO receive (chosen,
 135 rejected) pairs from the highest/lowest scoring responses. SP-RFT trains on the *chosen* trajectories
 136 from this same self-play dataset—not gold responses—so only the loss function differs between
 137 methods.¹ Because SP-RFT trains on model-generated chosen trajectories rather than gold or expert
 138 data, it is functionally *rejection sampling fine-tuning* (RFT) (Yuan et al., 2023); we use the SP-
 139 RFT label to distinguish it from standard SFT on gold data (Section 4.5). This distinction matters
 140 mechanistically: at small scale, where few model samples are correct, RFT on rare successes yields
 141 large gains; at large scale, where the model already solves most problems, RFT encounters a ceiling
 142 effect while DPO’s contrastive signal can still extract improvements (see Section 5).

143 **Design rationale.** Data is generated *at each scale*, matching how practitioners train (Shao et al.,
 144 2024; DeepSeek-AI, 2025; Chen et al., 2024). Dataset quality therefore correlates with model scale;
 145 we address this confound with a fixed-dataset control (Section 4.5). **Training.** AdamW, cosine LR
 146 (10^{-6}) with 10% warmup, 3 epochs, batch size 8. LoRA (Hu et al., 2022) ($r=16$, $\alpha=32$) at $\geq 7B$;
 147 full FT at smaller scales. BF16 on H100 80GB; $\beta=0.1$, SimPO $\gamma=0.5$. **Learning rate.** We sweep
 148 LR for SP-RFT and DPO at 1.5B, 3B, and 7B (Appendix M–O). The default 10^{-6} is optimal at
 149 7B; higher rates degrade performance monotonically, with DPO more sensitive than SP-RFT (DPO
 150 collapses to 16% at 10^{-5} vs. SP-RFT at 53%). The ranking inversion (DPO > SP-RFT) is preserved
 151 at independently optimized LR for each algorithm. At 1.5B, SP-RFT leads at every LR tested, with
 152 DPO collapsing at $\geq 5 \times 10^{-6}$.

¹**Effective dataset overlap.** A prompt enters the SP-RFT set if at least one of its 16 samples is correct, and enters the DPO/SimPO/IPO set if it has both a correct and an incorrect sample. The two sets coincide on the dominant central regime where pass@16 is neither 0 nor 1; they diverge only in the tails (very low or very high base accuracy). At 0.5B/1.5B/7B Instruct on Qwen 2.5 the asymmetric drops are small ($< 5\%$ of prompts) and affect both methods directionally similarly, leaving the inversion intact; per-scale effective-dataset sizes will be released with the per-seed outputs.

Table 1: GSM8K exact-match accuracy (%) across scales (Qwen 2.5). 1.5B: 3-seed means $\pm\sigma$; 7B: 5-seed means for SP-RFT/DPO/SimPO, 3-seed for IPO/KTO. Best per scale in **bold**. ^bSingle seed. ^cEpoch-2 checkpoint. ^dBest of 3 epochs. ^a62.9% flexible but 10.69% strict-match. **The 3B column reports the shared-default LR (10^{-6}) used at all other scales**; this matches the controlled-comparison protocol but is at the bottom of a hyperparameter pathology specific to the 3B Qwen 2.5 Instruct checkpoint. With per-algorithm best-LR tuning (Appendix O), SP-RFT reaches 55.7% at 10^{-5} and DPO 34.6% at 5×10^{-6} ; SP-RFT > DPO at *every* LR tested. The default-LR numbers are reported here to keep Table 1 as a controlled comparison. **0.5B column**: multi-seed reruns under the corrected sampler seeding (framework v1.9.0; Appendix K) — the four stable offline methods span only 0.79 pp, below the seed-noise floor, so no 0.5B ordering is statistically meaningful. **SGRPO at 3B** falls below base because on-policy rollouts receive zero reward for ill-formatted generations, so the format-compliance crisis compounds (Appendix J).

Algorithm	0.5B	1.5B	3B (full FT)	7B (LoRA)
Base Model	19.56	31.16	10.69 ^a	75.82
SP-RFT	33.59 $\pm$ 0.26 (+14.0)	54.36 $\pm$ 0.59 (+23.2)	18.35^b (+7.7)	77.38 $\pm$ 1.11 (+1.6)
DPO	33.18 $\pm$ 0.31 (+13.6)	49.08 $\pm$ 0.61 (+17.9)	14.44 $\pm$ 2.01 (+3.8)	83.38$\pm$0.56 (+7.6)
IPO	33.97 $\pm$ 0.68 (+14.4)	52.24 $\pm$ 0.22 (+21.1)	—	80.39 $\pm$ 0.92 (+4.6)
KTO	33.38 $\pm$ 0.98 (+13.8)	51.15 $\pm$ 1.77 (+20.0)	—	80.24 $\pm$ 0.16 (+4.4)
SimPO	23.55 $\pm$ 5.19 (+4.0)	38.67 $\pm$ 1.78 (+7.5)	—	83.32 $\pm$ 1.79 (+7.5)
SGRPO (GRPO)	32.45 ^b (+12.9)	58.00$\pm$0.57 (+26.8)	8.11 ^c (−2.6)	80.59 ^d (+4.8)

4 Results

4.1 Finding 1: Rankings Invert with Scale

Table 1 presents our main results on Qwen 2.5. At **0.5B**, five of six methods converge to identical strict-match accuracy across seeds ($\sigma=0$ for SP-RFT/DPO/KTO/SimPO/SGRPO; only IPO has $\sigma=0.26$). This is a *noise-floor* effect—0.5B is below the protocol’s differentiation threshold—rather than algorithmic equivalence; we retain it for completeness but the substantive scaling signal begins at 1.5B (Appendix K). At **1.5B**, SGRPO (online RL) tops all methods at $58.0\% \pm 0.57$. Among offline methods, SP-RFT leads ($54.4\% \pm 0.59$), followed by IPO, KTO, DPO, and SimPO. At **3B**, strict-match accuracy drops well below 1.5B—counter-intuitive but *not* a genuine scaling regression: the Qwen 2.5-3B-Instruct checkpoint has a pre-existing format-compliance deficit (62.9% flexible-match, 10.7% strict-match) that the shared default LR (10^{-6}) cannot overcome. A best-LR sweep (Appendix O) confirms this is a hyperparameter pathology: SP-RFT recovers to **55.7%**—*on par with 1.5B* (54.4%)—and SP-RFT > DPO holds at every LR tested. The 1.5B $\rightarrow$ 7B inversion claim is unaffected; 3B is a checkpoint-specific noisy interior point.

The ordering *inverts* at **7B** (Figure 2). With $N=5$ seeds for SP-RFT, DPO, and SimPO, DPO reaches $83.38\% \pm 0.56$ and SimPO $83.32\% \pm 1.79$; a Welch’s *t*-test fails to detect a significant difference ($p = 0.95$; note that a high *p*-value indicates failure to reject the null, not confirmation of equivalence—our sample size may lack power to resolve small gaps). The critical finding: SP-RFT drops from best offline method to *worst* at 7B ($77.38\% \pm 1.11$), separated by ~ 6 pp from DPO/SimPO ($p < 0.001$ after Bonferroni correction over the family of 10 pairwise tests at 7B; $\alpha = 0.005$, see Appendix K). This is the SP-RFT-to-preference inversion: the method that wins at small scale loses at large scale.

Disentangling scale from LoRA. The headline curve mixes full FT at $\leq 3B$ with LoRA at $\geq 7B$ —a practitioner-realistic choice (full FT is infeasible at 7B+ on common hardware) that confounds scale with training method. A 2×2 factorial at 7B trains SP-RFT/DPO/SimPO under *both* regimes: DPO leads under Full FT (84.5%) just as under LoRA (83.4%), and the inversion appears in both cells (Appendix D), so **scale**, not LoRA, drives the inversion; the mixed-method curve remains a residual limitation that full-FT runs at every scale could close.

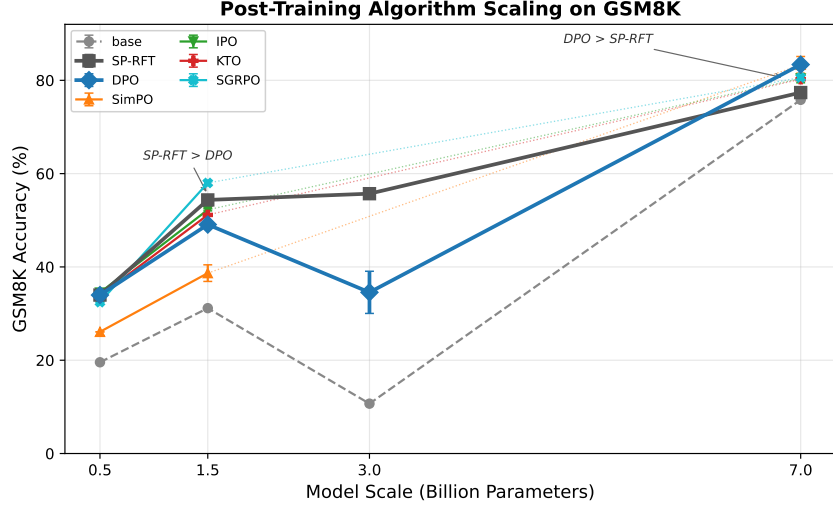


Figure 2: GSM8K accuracy across model scales (Qwen 2.5). At 7B, no significant difference is detected between DPO and SimPO ($p = 0.95$, $N=5$), while SP-RFT separates by ~ 6 pp ($p < 0.001$). The SP-RFT-to-preference **ranking inversion** between ≤ 3 B and 7B is the paper’s central finding.

4.2 Cross-Architecture Validation and Deployment Scale

Gemma 3 (1B–12B). A natural objection is that the ranking inversion may be a Qwen-specific artifact. To test generality, we train SP-RFT, DPO, and SimPO on Gemma 3 IT (Gemma Team, 2025) at 1B, 4B, and 12B (3 seeds each, 27 runs). The pattern mirrors Qwen. At **1B**, SP-RFT dominates (26.7%), DPO falls to 21.5%, and SimPO collapses to 4.1%. At **4B**, the same ordering holds with wider gaps: SP-RFT (75.30 ± 0.53) $\gg$ DPO (57.77 ± 10.76) $\gg$ SimPO (40.03 ± 8.31); Welch’s t gives SP-RFT $>$ DPO at $p=0.11$ (limited power at $N=3$ with high DPO variance) and SP-RFT $>$ SimPO at $p=0.018$. At **12B**—the critical test—DPO (87.4%) overtakes SP-RFT (86.7%), with Welch’s t giving DPO $>$ SP-RFT at $p=0.036$ ($N=3$, uncorrected); the 0.7 pp gap is small but the directional flip is clear and replicates the Qwen pattern. SimPO (75.1%) remains behind ($p=0.048$ vs. DPO).

Two differences from Qwen are worth noting. First, SimPO does not converge to DPO at the largest scale in Gemma, suggesting that the DPO/SimPO parity observed at Qwen2.5-7B may be architecture-specific. Second, the inversion point appears later in Gemma (12B vs. 7B), which may reflect differences in the pretrained checkpoints’ answer-format compliance under strict-match grading (Appendix H). Despite these nuances, the central finding—that SP-RFT-to-preference rankings invert with scale—replicates across architectures.

Qwen2.5-14B. To extend the scaling curve beyond 12B, we train SP-RFT, DPO, and SimPO on Qwen 2.5-14B-Instruct with 3 seeds each (9 runs). The inversion persists: DPO ($86.8\% \pm 0.23$) $>$ SimPO ($86.5\% \pm 0.68$) $>$ SP-RFT ($84.5\% \pm 0.19$) $>$ Base (79.5%). Welch’s t -tests at $N=3$: DPO $>$ SP-RFT significant ($p=0.0002$); SimPO $>$ SP-RFT marginally significant ($p=0.030$); SimPO vs. DPO not detectable ($p=0.53$, gap 0.30 pp $\ll$ seed noise). Within the Qwen family the DPO–SP-RFT gap *narrows* from 6.0 pp at 7B to **2.2 pp** at 14B; on general benchmarks (ARC-C, HellaSwag, WinoGrande) the spread is < 1.2 pp.

Notably, MATH evaluation at 14B reveals a different pattern: SimPO (39.1%) dramatically outperforms DPO (25.8%) and SP-RFT (23.0%), contrasting with Gemma 12B where DPO dominates MATH (50.2% vs. SimPO 32.7%). This architecture-dependent reversal on harder tasks further underscores that algorithm rankings cannot be extrapolated across models or tasks.

Table 2 presents the combined cross-architecture and deployment-scale results.

Table 2: Cross-architecture validation and deployment scale on GSM8K (%). Gemma 3: 3 seeds each; 12B uses LoRA. Qwen2.5-14B: 3 seeds each, LoRA. The SP-RFT-to-DPO inversion replicates across architectures and persists at 14B.

Algorithm	Gemma 1B		Gemma 12B		Qwen2.5-14B	
	Mean	$\pm\sigma$	Mean	$\pm\sigma$	Mean	$\pm\sigma$
SP-RFT	26.7	—	86.7	0.29	84.5	0.19
DPO	21.5	—	87.4	0.26	86.8	0.23
SimPO	4.1	—	75.1	4.88	86.5	0.68
<i>SP-RFT–DPO gap</i>	<i>+5.2 pp</i>		<i>−0.7 pp</i>		<i>−2.2 pp</i>	

Table 3: Base vs. Instruct initialization at 1.5B on GSM8K (%). The spread collapses from 15.7 pp to 1.07 pp; SimPO reverses from worst to best. This compression generalizes to 7B (6.00 pp $\rightarrow$ 1.82 pp; Appendix Q). **Caveat:** Base-init values are single-seed point estimates; the 1.07 pp spread should therefore be read as suggestive (within plausible seed noise of $\sigma \sim 0.6$ observed for 1.5B Instruct). The *direction* of the compression (Instruct $\gg$ Base spread) is much larger than seed noise even on a generous reading.

Algorithm	From Base	From Instruct	Δ
SP-RFT	60.80	54.36 ± 0.59	+6.4
DPO	61.33	49.08 ± 0.61	+12.2
SimPO	61.87	38.67 ± 1.78	+23.2
<i>Spread</i>	<i>1.07 pp</i>	<i>15.7 pp</i>	

209 4.3 Finding 2: Initialization Dominates Algorithm Choice

210 Why do rankings invert? We hypothesize that algorithms interact differently with the prior alignment
211 baked into Instruct checkpoints, and that removing this alignment should eliminate the differences.

212 Table 3 tests this at 1.5B. Starting from Base, all methods improve (SP-RFT: +6.4 pp, DPO: +12.2 pp,
213 SimPO: +23.2 pp) and the spread collapses from 15.7 pp to just **1.07 pp**. SimPO—worst from
214 Instruct—becomes best from Base, suggesting its reference-free nature makes it most sensitive to
215 initialization.

216 Critically, this finding generalizes to 7B (Appendix Q): from Base, the spread collapses from 6.00 pp
217 to 1.82 pp. The ranking from Base does *not* simply reproduce the 1.5B pattern: at 1.5B Base,
218 SimPO leads (61.87%) followed by DPO (61.33%) and SP-RFT (60.80%); at 7B Base, SP-RFT leads
219 (82.49%) followed by SimPO (80.82%) and DPO (80.67%). Thus, a residual ranking shift persists
220 even from Base—but the key observation is that these differences are compressed to ≤ 1.82 pp,
221 compared to 6–16 pp from Instruct. Initialization is the dominant source of inter-algorithm variance,
222 even if it does not fully explain it.

223 DPO’s sensitivity to Base initialization is theoretically expected (KL regularization anchors to a
224 reference that cannot format math), but the convergence extends to reference-free methods (SimPO)
225 and pure SP-RFT, so the prior alignment state—not just reference-model mechanics—is the primary
226 source of inter-algorithm variance. Practically, most real-world post-training starts from Instruct,
227 where algorithm choice does produce measurable differences; the Base ablations at both scales reveal
228 that those differences are largely an initialization-loss interaction.

229 4.4 Finding 3: Two Regimes—Format Compliance, Then Architecture-Specific Reasoning

230 Tables 4 and 5 compare GSM8K and MATH results. At 1.5B, the 19.3 pp GSM8K spread collapses to
231 **0.54 pp** on MATH—a $36\times$ compression. Rankings shift: SGRPO drops from 1st to 4th, SimPO rises
232 from last to 2nd. On general-domain benchmarks (ARC-C, HellaSwag, WinoGrande), the spread
233 collapses further to **0.47 pp** ($41\times$; Appendix G). At small scale, algorithms differ only in format
234 compliance, not reasoning (Table 4; Appendix H).

Table 4: Cross-task comparison at 1.5B: GSM8K vs. MATH. The 19.3 pp spread collapses to 0.54 pp ($36\times$ compression).

Algorithm	MATH	GSM8K
SP-RFT	27.04	54.36
SimPO	26.72	38.67
KTO	26.64	51.15
SGRPO	26.58	58.00
DPO	26.50	49.08
<i>Spread</i>	<i>0.54</i>	<i>19.33</i>

Table 5: MATH accuracy (%) across scales. At $\geq 4B$, DPO leads in Gemma but SimPO leads in Qwen2.5-14B.

	Gemma 3			Qwen
	1B	4B	12B	14B
Base	16.6	34.4	41.3	16.8
SP-RFT	13.7	34.3	41.0	23.0
DPO	16.4	42.4	50.2	25.8
SimPO	1.1	39.2	32.7	39.1

Table 6: Fixed-dataset control: 1.5B on 14B self-play data (GSM8K %). SP-RFT still leads by +5.1 pp.

Algorithm	Mean	σ
SP-RFT	58.38	0.49
DPO	53.32	0.70
<i>Gap</i>	<i>+5.06 pp</i>	

Table 7: Gold-SFT control: standard SFT on MetaMathQA (GSM8K %). SFT advantage collapses from +12 pp to ≈ 0 pp with scale.

Method	1.5B		7B	
	Mean	σ	Mean	σ
Gold-SFT	61.05	0.45	83.32	0.28
SP-RFT	54.36	0.59	77.38	1.11
DPO	49.08	0.61	83.38	0.56

235 **But this picture changes at scale.** Table 5 shows MATH accuracy across Gemma 3 scales. At
 236 1B, the raw spread is 15.5 pp, but this is driven entirely by SimPO’s catastrophic collapse to 1.1%;
 237 excluding SimPO, Base/SP-RFT/DPO cluster within 2.9 pp. At 4B, DPO pulls ahead by +8.1 pp
 238 over SP-RFT. At 12B, DPO reaches 50.2%—nearly **+9 pp over base and +17 pp over SimPO**.
 239 The spread among non-pathological methods *increases* with scale, the opposite of the compression
 240 observed at 1.5B.

241 This pattern is *architecture-dependent*: at Qwen2.5-14B, **SimPO** dominates MATH ($39.1\% \pm 2.2$)
 242 over DPO ($25.8\% \pm 0.4$) by +13 pp—the reverse of Gemma. Meanwhile general benchmarks
 243 (ARC-C, HellaSwag, WinoGrande; Appendix T) show <2 pp spread at all scales/architectures, but
 244 those are multiple-choice and do not test open-ended generation, so they do not rule out inversions
 245 on generative tasks (AlpacaEval, MT-Bench). Two-regime picture: at small scale algorithms differ
 246 shallowly (format); at large scale they differ deeply (reasoning), but *which algorithm wins depends*
 247 *on architecture and task*—both regimes make small-scale extrapolation unreliable.

248 4.5 Controls: Disentangling Data Quality and RFT Artifacts

249 **Fixed-dataset control.** To disentangle data quality from model scale, we train 1.5B Qwen 2.5 Instruct
 250 on self-play data generated by the 14B model. **SP-RFT still beats DPO by +5.1 pp** ($58.38\% \pm 0.49$
 251 vs. $53.32\% \pm 0.70$, 3 seeds each); both algorithms gain comparably from higher-quality data (SP-RFT:
 252 +4.0 pp, DPO: +4.2 pp), leaving the gap stable (5.06 vs. 5.28 pp with own data). Model scale, not
 253 data quality, is the primary driver.

254 **Gold-data SFT control.** To rule out an RFT-ceiling artifact, we train standard SFT on Meta-
 255 MathQA (Yu et al., 2024) (240K external expert solutions; 3 seeds at both 1.5B and 7B). Tables 6–7
 256 report both controls side by side: at **1.5B**, Gold-SFT leads at $61.05\% \pm 0.45$ (+12.0 pp over DPO);
 257 at **7B**, Gold-SFT converges to DPO ($83.32\% \pm 0.28$ vs. $83.38\% \pm 0.56$). The SFT advantage col-
 258 lapses from +12 pp to ≈ 0 pp with scale regardless of whether the training data is self-generated or
 259 external—**falsifying the RFT-artifact hypothesis**.

260 4.6 Algorithm Selection Under Scale Shift

261 Given that small-scale rankings are anti-predictive, what should a practitioner do when target-scale
 262 validation is too expensive? Over the cross-scale ranks in Tables 1–2 we report two summary criteria:
 263 *worst-case rank* $R_{wc} = \max_N r_a^{(N)}$ and *deployment rank* $R_{dep} = \text{mean}_{N \geq 7B} r_a^{(N)}$. Values: SP-RFT
 264 ($R_{wc}=4, R_{dep}=3.5$), DPO (4, 1), SimPO (5, 2). DPO Pareto-dominates SP-RFT on R_{dep} at equal
 265 R_{wc} , and dominates SimPO on both; Gemma 3 replicates the order. **DPO is therefore the rank-**
 266 **stable bet**—best from 7B onward across both architectures, lowest seed variance ($\sigma=0.56$ at 7B
 267 vs. SimPO’s 1.79). Caveat: at 14B DPO (86.78 ± 0.23) and SimPO (86.48 ± 0.68) overlap within
 268 seed noise, so R_{dep} should be read with bootstrap rank intervals (Appendix G) rather than as a hard
 269 ordering. Empirically the DPO–SP-RFT gap turns positive past an architecture-specific crossover
 270 (Qwen between 3B and 7B; Gemma between 4B and 12B), so use DPO when deployment scale
 271 is within $2\times$ of that crossover; we avoid a parametric $\Delta(N)$ form because the within-Qwen gap
 272 narrows from +6.0 pp at 7B to +2.2 pp at 14B. The mechanistic basis (Section 5) is that SP-RFT hits
 273 a ceiling once base accuracy is high, while contrastive objectives still extract gradient from incorrect
 274 trajectories.

275 4.7 Practitioner Recommendations

276 Actionable guidance: **(1) Validate at deployment scale, architecture, and task**—rankings at $\leq 3B$
 277 do not predict 7B+ behavior, and MATH rankings flip between Gemma 12B (DPO) and Qwen2.5-14B
 278 (SimPO). **(2) DPO is a strong default on GSM8K**—near top at 1.5B/7B Qwen, overtakes SP-RFT
 279 at 12B Gemma and 14B Qwen, lowest seed variance ($\sigma=0.56$ at 7B), but does not transfer uniformly
 280 to MATH. **(3) Distrust variant sweeps at small scale**—the 1.5B→7B variant ranking fully inverts;
 281 prefer vanilla DPO unless you can validate at target scale. **(4) Budget for multiple seeds**—SimPO
 282 matches DPO in mean at 7B but with $3.2\times$ the variance.

283 5 Discussion

284 **Why rankings invert: a mechanistic sketch.** The Base-vs-Instruct ablation (Section 4.3) identifies
 285 the core mechanism: at small scale the Instruct model rarely produces correct solutions, so RFT yields
 286 large gains while preference methods waste capacity on pairs of incorrect outputs; at 7B RFT hits a
 287 ceiling while DPO’s contrastive signal still extracts gradient from incorrect trajectories. Gold-data
 288 SFT (MetaMathQA) also converges to DPO at 7B (83.32 vs. 83.38) despite leading by +12 pp at
 289 1.5B, ruling out self-play data saturation as the cause.

290 **What drives accuracy in mathematical reasoning.** In decreasing order of leverage: model scale
 291 (~ 50 pp), initialization (~ 15 pp), training paradigm (SP-RFT vs. preference: ~ 6 pp at 7B), and
 292 loss function choice within the preference family (< 1 pp for DPO vs. SimPO on GSM8K; up to
 293 5.8 pp across DPO variants at 7B). On harder reasoning tasks (MATH), a new dimension emerges:
 294 algorithm-architecture interaction. DPO gains +9 pp on Gemma 12B MATH while SimPO gains
 295 +13 pp on Qwen2.5-14B MATH—a complete reversal. This suggests that loss functions interact with
 296 architecture-specific representations in ways that are invisible on format-sensitive tasks like GSM8K.

297 **Methodological caveats at 7B+.** LR 10^{-6} is below the standard LoRA recommendation but ablations
 298 rule out under-training (Appendix M); SGRPO without explicit KL loses only 0.15 pp (Appendix P);
 299 full-FT/LoRA factorial (Appendix D) confirms the inversion in both regimes. Online RL ablations in
 300 Appendix E.

301 6 Conclusion

302 On mathematical reasoning, post-training algorithm rankings invert with scale: the method that
 303 wins at 1.5B loses at 7B, replicated across Qwen 2.5 (up to 14B) and Gemma 3 (up to 12B),
 304 confirmed with $p < 0.001$ at $N=5$, and surviving fixed-dataset and gold-data SFT controls. On MC
 305 general benchmarks the spread is < 1.2 pp at all scales—the inversion is task-specific. Initialization,

not loss function, explains most algorithmic differences ($3\text{--}15\times$ spread compression from Base). The dominant paradigm of small-scale benchmarking + extrapolation is therefore unreliable for mathematical reasoning; practitioners should validate at deployment scale, and the community should invest in how algorithms interact with scale, initialization, and task structure rather than designing new losses at a single small scale.

Limitations. Two families on math only; MC-only general benchmarks limit transfer to generative tasks. Fixed-dataset control is asymmetric (no 7B-on-1.5B-data run); 0.5B and 3B Qwen have noise-floor and LR-pathology caveats (Appendices K,O). Code and data release upon acceptance.

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A Full DPO Variant Results

Statistical notes. Significance threshold: Bonferroni-corrected $\alpha = 0.05/19 = 0.0026$. Only SimPO ($p < 10^{-4}$) passes, and it is significantly *worse*. ORPO ($p = 0.013$) and DPOP ($p = 0.019$) would be significant uncorrected but fail after correction. Our 5-seed design provides 80% power to detect effects ≥ 1.0 pp (Cohen’s $d \geq 0.8$).

At 7B: point-estimate rankings reverse. We evaluate a stratified subset of 5 variants at 7B (3 seeds each), selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th). ORPO—1st at 1.5B—collapses

Table 8: Taxonomy of 20 DPO variants evaluated in this study, organized by primary modification to the DPO objective. All evaluated at 1.5B on GSM8K with 5 seeds.

Variant	Primary Modification	Ref.	Year
DPO	Baseline logistic loss	Y	2023
IPO	Squared loss regularization	Y	2024
SimPO	Length-normalized, no ref	N	2024
KTO	Unpaired binary feedback	Y	2024
Hinge	Hinge loss replacement	Y	2024
CPO	Contrastive preference	N	2024
ORPO	SFT-integrated preference	N	2024
RDPO	Robust divergence penalty	Y	2024
CDPO	Conservative DPO	Y	2024
BetaDPO	Adaptive β scheduling	Y	2024
CalDPO	Calibrated reward margin	Y	2024
DPOP	Pref.-weighted optimization	Y	2024
ODPO	Online DPO	Y	2024
EXO	Exponentiated gradient	Y	2024
AlphaPO	Alpha-divergence pref.	N	2024
APO	Anchored preference	Y	2024
SPPO	Self-play preference	Y	2024
RobustDPO	Dist. robust DPO	Y	2024
GPO	Generalized pref. opt.	Y	2024
FocalPO	Focal loss preference	Y	2024

Table 9: Full DPO variant results at 1.5B scale on GSM8K (5 seeds). Δ is the accuracy difference from vanilla DPO. † indicates statistical significance after Bonferroni correction ($p < 0.0026$). Cohen’s d measures practical effect size.

Variant	Category	Mean (%)	Std	Δ DPO	p	d
ORPO	Reference-free	53.89	0.60	+4.12	0.0134	2.48
DPOP	Weighting/calibration	53.62	1.80	+3.85	0.0189	1.88
SPPO	Data augmentation	52.60	1.45	+2.84	0.0521	1.49
IPO	Alternative divergence	52.36	0.75	+2.59	0.0614	1.53
CalDPO	Weighting/calibration	52.16	1.58	+2.40	0.0934	1.22
CPO	Reference-free	51.43	1.57	+1.67	0.2187	0.85
GPO	Alternative divergence	50.98	2.41	+1.21	0.4370	0.52
RobustDPO	Robustness	50.89	2.13	+1.12	0.4442	0.51
CDPO	Robustness	50.72	1.89	+0.96	0.4916	0.46
KTO	Unpaired	50.48	2.11	+0.71	0.6211	0.33
Hinge	Alternative divergence	49.87	2.43	+0.11	0.9449	0.05
DPO	Vanilla	49.76	2.27	—	—	—
ODPO	Data augmentation	48.37	3.20	−1.39	0.4525	−0.50
APO	Data augmentation	48.07	2.73	−1.70	0.3175	−0.68
EXO	Data augmentation	47.52	2.92	−2.24	0.2145	−0.86
FocalPO	Weighting/calibration	47.04	3.27	−2.73	0.1686	−0.97
RDPO	Alternative divergence	46.79	2.47	−2.97	0.0835	−1.25
BetaDPO	Weighting/calibration	45.75	4.31	−4.02	0.1145	−1.17
AlphaPO	Reference-free	41.68	5.13	−8.08	0.0204	−2.04
SimPO †	Reference-free	38.23	2.12	−11.54	0.0000	−5.25

to worst at 7B (−4.79 pp vs. DPO). EXO—15th at 1.5B—rises to the top (+1.00 pp). No variant reaches Bonferroni-corrected significance at either scale. The practical conclusion: since variant rankings are statistically unresolvable at both scales and their point estimates reverse, small-scale variant sweeps provide no reliable signal for large-scale behavior.

Shared-hyperparameter design. All 20 variants use the same LR (10^{-6}) and $\beta = 0.1$. This is a deliberate choice: the 20-variant sweep evaluates *default-hyperparameter transferability*—whether variants provide gains “out of the box” with shared settings—not absolute algorithmic capability ceilings. This matches how practitioners typically deploy variants: using the original DPO paper’s

defaults. Per-variant tuning (e.g., optimizing β for IPO’s squared loss or KTO’s unpaired feedback) could reveal differences that our shared-hyperparameter design masks. Our conclusion (variants are unresolvable) applies to the plug-and-play setting; per-variant tuning could surface real differences that shared hyperparameters mask.

Table 10: Stratified DPO variant evaluation at 7B (3 seeds each). Variants were selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th) to test rank-order predictiveness. Vanilla DPO: 83.38% ($N=5$). Point-estimate rankings reverse: the 1.5B leader (ORPO) is the 7B laggard; the 1.5B underperformer (EXO) rises to top. No variant reaches Bonferroni-corrected significance at either scale.

Variant	Mean (%)	Std	Δ DPO	p	1.5B Rank
EXO	84.38	0.96	+1.00	0.277	15th
ODPO	83.88	1.31	+0.49	0.654	13th
GPO	83.40	1.05	+0.02	0.984	7th
Vanilla DPO	83.38	0.56	—	—	12th
DPOP	80.92	1.18	−2.46	0.085	2nd
ORPO	78.59	1.04	−4.79	0.013	1st

B Training Configurations

Table 11: Shared training configuration across all experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Weight decay	0.01
LR schedule	Cosine with 10% linear warmup
Effective batch size	8 prompts (1×8 grad. accum.)
Training epochs	3
Precision	BF16 mixed precision
Distributed strategy	DeepSpeed ZeRO Stage 3
Hardware	$8 \times$ NVIDIA H100 80GB HBM3
Sequence length	512 (math), 1024 (code)

Table 12: Algorithm-specific hyperparameters.

Algorithm	Learning Rate	Key Hyperparameters	GPUs/Run
SP-RFT	1×10^{-6}	—	1
DPO	1×10^{-6}	$\beta = 0.1$	1
SimPO	1×10^{-6}	$\beta = 0.1$, $\gamma = 0.5$	1
IPO	1×10^{-6}	$\beta = 0.1$	1
KTO	1×10^{-6}	$\beta = 0.1$	1
SGRPO	1×10^{-6}	$\lambda_{\text{KL}} = 0^*$, $n = 4$	2
GSPO	1×10^{-6}	$\lambda_{\text{KL}} = 0$, $n = 4$	2
CISPO	1×10^{-6}	$\lambda_{\text{KL}} = 0$, $n = 4$, clip = $[0.8, 1.2]$	2

recipe. See Appendix P for an ablation with $\lambda_{\text{KL}} = 0.01$.

For models at 7B/12B and for the 3B factorial, we apply LoRA (Hu et al., 2022) with rank $r = 16$, $\alpha = 32$, applied to all linear layers. The core comparison at $\leq 3\text{B}/4\text{B}$ uses full fine-tuning.

C Framework Engineering

Scaling online RL to 7B required resolving three critical infrastructure issues: (1) **Ray GPU over-subscription**. Fractional num_gpus values in Ray caused GPU over-subscription deadlocks during multi-worker rollout generation; we enforce integer GPU allocation per worker. (2) **vLLM memory**

411 **leak.** Rollout output objects were not explicitly freed between batches, causing a slow memory
412 leak that triggered OOM after ~ 200 steps at 7B; explicit deletion with `gc.collect()` resolved
413 this. (3) **Epoch-transition OOM.** Optimizer state accumulated across epoch boundaries; offloading
414 optimizer state to CPU between epochs eliminated the failure.

415 D LoRA Factorial

Table 13: 2×2 factorial disentangling scale from LoRA (Qwen 2.5). The inversion appears under both Full FT and LoRA at 7B: DPO (84.5%) > SimPO (83.6%) > SP-RFT (79.8%) under Full FT, mirroring the LoRA ranking. 7B LoRA values: 5-seed means $\pm \sigma$; Full FT: single seed.

		3B	7B
Full FT	Base	10.69	75.82
	SP-RFT	18.35^b (+7.7)	79.83 (+4.0)
	DPO	14.44 ± 2.01 (+3.8)	84.46 (+8.6)
	SimPO	6.90 ^b (−3.8)	83.62 (+7.8)
LoRA	Base	10.69	75.82
	SP-RFT	18.04 (+7.4)	77.38 ± 1.11 (+1.6)
	DPO	16.00 (+5.3)	83.38 ± 0.56 (+7.6)
	SimPO	6.90 (−3.8)	83.32 ± 1.79 (+7.5)

416 E Online RL Explorations

417 SGRPO (online RL) leads at 1.5B (58.0% $\pm$ 0.57, +3.6 pp over SP-RFT) at $\sim 6\times$ the compute cost.
418 At 7B, SGRPO achieves 80.59% (best of 3 epochs), placing it above SP-RFT but below DPO/SimPO.
419 Adding KL regularization ($\lambda_{\text{KL}} = 0.01$) yields only +0.15 pp (Appendix P), confirming that the
420 online–offline gap is genuine rather than a regularization artifact. **Caveat:** SGRPO uses the same
421 effective batch size (8 prompts) and fixed 3-epoch schedule as offline methods; online RL typically
422 benefits from larger batches and dynamic rollout budgets, so our SGRPO results may underestimate
423 online RL’s true ceiling. SGRPO was not evaluated at 12B or 14B due to its $\sim 6\times$ compute cost.
424 Token-level SGRPO outperforms sentence-level GSPO by 12–18 pp; GSPO and CISPO results are
425 below.²

426 **SGRPO at 3B.** SGRPO at 3B achieves 4.93% after epoch 1 and 8.11% after epoch 2, both below
427 the 10.69% baseline. On-policy rollouts compound the format compliance crisis: poorly formatted
428 generations receive no reward, preventing the policy from learning formatting.

429 **SGRPO at 7B.** SGRPO achieves 80.59% (best of 3 epochs; 80.59/80.29/79.91), above SP-RFT but
430 below DPO/SimPO. Slight accuracy degradation at epochs 2–3 suggests overfitting to the reward
431 signal.

432 **GSPO and CISPO — caveat: implementation status not externally validated.** GSPO (sentence-
433 level importance ratio) achieves 39.58% at 1.5B and 19.86% at 0.5B—12–18 pp below token-level
434 SGRPO. CISPO (clipped ratio as a detached weight, clip [0.8, 1.2]) fails catastrophically (0.15%
435 at 0.5B, 5.23% at 1.5B). **We do not yet have an externally-published reference number for
436 either GSPO or CISPO on Qwen 2.5 with our exact training recipe, so we cannot rule out
437 implementation issues**—the most likely candidate is rollout/trainer log-probability drift in the
438 sequence-level estimator (which GSPO is designed to absorb but CISPO is more sensitive to).
439 We therefore report these numbers as observed under our framework but do not draw method-level
440 conclusions from them; the main paper’s claims rely on the offline methods and SGRPO. Re-validating
441 GSPO/CISPO against an external benchmark is on our open-question list.

²GSPO achieves 39.58% at 1.5B; CISPO fails catastrophically (5.23% at 1.5B, 0.15% at 0.5B).

F SimPO Variance Analysis

SimPO exhibits the highest seed variance of any method at any scale ($\sigma = 1.79$ at 7B vs. DPO’s $\sigma = 0.56$). Across 5 seeds, SimPO’s per-seed accuracy ranges from 81.50% to 85.82% (range: 4.32 pp), while DPO ranges from 82.79% to 84.23% (range: 1.44 pp)—a $3\times$ difference. SimPO’s two highest and two lowest seeds cluster in pairs, consistent with a bifurcation in the optimization landscape. This bimodal pattern is absent in DPO. SimPO computes length-normalized log-probabilities without a reference model, lacking the implicit regularization of DPO’s reference-anchored gradient. The practical consequence: SimPO’s “best seed” appears to outperform DPO (85.82 vs. 84.23), but no significant difference is detected under proper multi-seed evaluation ($p = 0.95$).

G Additional Figures and Tables

Table 14: General-domain evaluation at 1.5B: ARC-Challenge (acc_norm), HellaSwag (acc_norm), WinoGrande (acc). The 19.3 pp GSM8K spread collapses to 0.47 pp ($41\times$ compression).

Algorithm	ARC-C (%)	HellaSwag (%)	WinoGrande (%)	Avg (%)
Base Model	46.76	68.23	62.90	59.30
SP-RFT	46.16	68.16	62.90	59.08
DPO	46.67	68.38	62.67	59.24
IPO	46.76	68.36	62.67	59.26
KTO	47.01	68.25	63.06	59.44
SimPO	46.08	68.37	62.59	59.01
SGRPO	46.67	68.23	63.54	59.48
<i>Spread</i>	<i>0.93</i>	<i>0.22</i>	<i>0.95</i>	<i>0.47</i>

Table 15: General-domain evaluation at 7B (LoRA). Despite a 6.0 pp GSM8K spread, the general-domain spread is just 0.71 pp.

Algorithm	ARC-C (%)	HellaSwag (%)	WinoGrande (%)	Avg (%)
Base Model	55.20	80.48	71.11	68.93
SP-RFT	55.29 (+0.09)	80.45 (−0.03)	70.72 (−0.39)	68.82 (−0.11)
DPO	54.86 (−0.34)	80.58 (+0.10)	70.72 (−0.39)	68.72 (−0.21)
IPO	55.55 (+0.35)	80.37 (−0.11)	71.35 (+0.24)	69.09 (+0.16)
KTO	55.03 (−0.17)	80.39 (−0.09)	70.72 (−0.39)	68.71 (−0.22)
SimPO	54.44 (−0.77)	80.53 (+0.05)	70.17 (−0.95)	68.38 (−0.56)
<i>Spread (trained)</i>	<i>1.11 pp</i>	<i>0.21 pp</i>	<i>1.18 pp</i>	<i>0.71 pp</i>

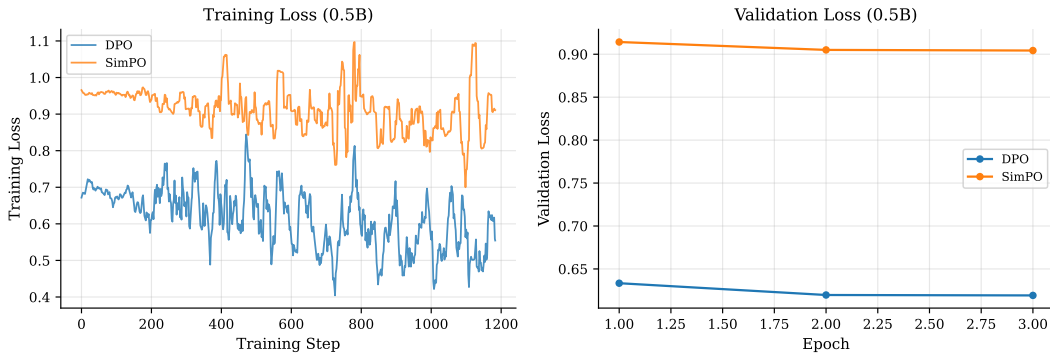


Figure 3: Training dynamics comparison at 0.5B scale. **Left:** training loss over steps (smoothed). **Right:** validation loss over epochs.

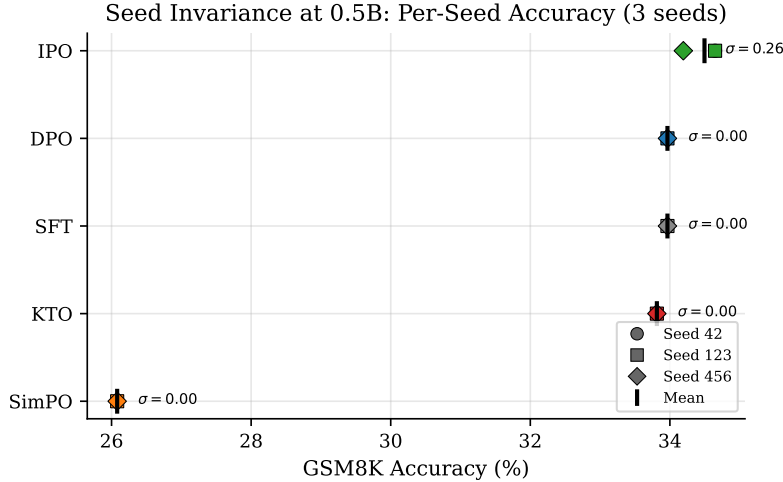


Figure 4: **0.5B is at the noise floor of our protocol, not a meaningful inter-algorithm comparison.** Per-seed GSM8K strict-match accuracy for each algorithm across 3 seeds. Five of six methods converge to identical accuracy ($\sigma=0$, all 33.81–33.97%); IPO is the only algorithm with non-trivial seed variance ($\sigma=0.26$). With greedy decoding and a 0.5B base that already scores 19.6% before training, post-training adds the same ~ 14 pp lift regardless of loss. We retain 0.5B in the scaling plot for completeness, but *the 0.5B column should not be read as evidence that algorithms are equivalent*—it is evidence that 0.5B Qwen 2.5 Instruct on GSM8K is below the differentiation threshold of any of these methods. The scaling-study signal therefore begins at 1.5B.

H Format Compliance Analysis

Table 16 reports strict-match and flexible-extract accuracy. The *format compliance gap* (strict – flexible) quantifies whether a model produces well-formatted answers.

Table 16: Strict-match vs. flexible-extract accuracy on GSM8K. At 3B, the format compliance gap ranges from -26.2 to -33.7 pp across trained models, confirming format non-compliance drives the low strict-match scores.

Scale	Algorithm	Strict (%)	Flexible (%)	Gap (pp)
1.5B	SP-RFT	54.66	55.57	−0.9
	DPO	51.18	49.81	+1.4
	IPO	52.39	52.31	+0.1
	KTO	51.71	51.18	+0.5
	SimPO	43.44	43.06	+0.4
3B (full FT)	Base	10.69	62.93	−52.2
	SP-RFT	18.04	44.28	−26.2
	DPO	16.45	42.91	−26.5
	IPO	16.83	45.72	−28.9
	KTO	15.31	44.50	−29.2
	SimPO	6.90	40.56	−33.7
7B (LoRA)	SP-RFT	76.42	80.29	−3.9
	DPO	83.85	84.99	−1.1
	IPO	81.35	83.09	−1.7
	KTO	80.21	82.49	−2.3
	SimPO	85.82	80.97	+4.9

Table 17: Compute budget breakdown on H100 80GB GPUs.

Category	# Runs	Time/Run	GPUs	GPU-h
Core 0.5B SL (5 algs $\times$ 3 seeds)	15	4–8 min	1	1.5
Core 0.5B RL (3 algs $\times$ 3 seeds)	9	$\sim$ 3 h	2	54
Core 1.5B SL (5 algs $\times$ 3 seeds)	15	14–27 min	1	5.0
Core 3B SL full FT (3 algs)	3	12–13 min	1	0.6
3B LoRA factorial (3 algs)	3	12–13 min	1	0.6
Core 7B SL (5 algs $\times$ 1 seed)	5	335–460 min	1	30
Core 7B RL (SGRPO, 3 epochs)	3	$\sim$ 480 min	2	48
7B multi-seed: SP-RFT/DPO/SimPO ($N=5$)	13	330–360 min	1	78
7B multi-seed: IPO/KTO ($N=3$)	4	335–460 min	1	27
MATH 1.5B evals (5 methods)	5	$\sim$ 30 min	1	2.5
DPO variants 1.5B (20 $\times$ 5 seeds)	100	14–27 min	1	30
DPO variants 7B (5 $\times$ 3 seeds)	15	330–360 min	1	90
Ablations 1.5B	$\sim$ 57	9–10 min	1	9
Ablations 7B (LR sweep, KL, Base init)	16	330–480 min	1–2	$\sim$ 100
Gemma 3 (3 algs $\times$ 3 scales $\times$ 3 seeds)	27	varies	1–2	$\sim$ 75
7B Full FT factorial (SP-RFT/DPO/SimPO)	3	$\sim$ 180 min	2	18
14B LoRA (SP-RFT/DPO/SimPO $\times$ 3 seeds)	9	$\sim$ 360 min	2	108
Fixed-dataset control (SP-RFT/DPO $\times$ 3 seeds)	6	14–27 min	1	2.0
1.5B LR sweep (SP-RFT/DPO $\times$ 4 LR $\times$ 3 seeds)	18	14–27 min	1	$\sim$ 5.0
3B LR sweep (SP-RFT/DPO $\times$ 3 LR $\times$ 3 seeds)	18	$\sim$ 25 min	1	$\sim$ 7.5
Gold-SFT control (1.5B $\times$ 3 + 7B $\times$ 3 seeds)	6	27–360 min	1	$\sim$ 20
Total training	$\sim$ 350	–	–	$\sim$720
Evaluation (GSM8K, MATH, general)	$\sim$ 430+	5–600 min	1	$\sim$ 220
Grand total				$\sim$940 GPU-h

I Compute Budget

The total compute ($\sim$ 940 GPU-hours) is modest given the breadth ($\sim$ 350 training runs across 2 model families, 8 scales, 4 evaluation domains) and substantially less than training a single 7B model from scratch.

J Additional Scaling Results

SGRPO at 1.5B achieves $58.00\% \pm 0.57$ (seeds: 57.47%, 57.92%, 58.61%). SL methods at 1.5B use multi-seed reruns with genuine variance (Table 19).

K Multi-Seed Verification

Why early runs were bit-identical across seeds. In framework versions before v1.9.0, the training-data `DistributedSampler` received a fixed default seed, so offline runs differing only in `run.seed` saw identical data order; with deterministic bf16 training and greedy evaluation, they were bit-reproducible. This produced the $\sigma=0$ cells originally reported at 0.5B. The sampler now consumes `run.seed`; the 1.5B and 7B tables below use post-fix reruns, and Table 18 reports a post-fix rerun of the *entire* 0.5B grid.

Statistical tests at 7B. Welch’s t -tests with Bonferroni correction ($\alpha = 0.05/10 = 0.005$):

- **DPO vs. SimPO:** $\Delta = +0.06$ pp, $p = 0.950$. Not significant; no difference detected at this sample size.

Table 18: 0.5B multi-seed rerun under corrected sampler seeding (GSM8K 0-shot CoT strict-match, 3 seeds each). Variance is genuine and nonzero; original point estimates were accurate to <1 pp; the four stable methods span 0.79 pp (below seed noise), and SimPO’s $\sigma=5.19$ replicates its instability (Appendix F) at a third scale.

Method	Mean $\pm\sigma$	s42	s123	s456
IPO	33.97 $\pm$ 0.68	33.97	33.28	34.65
SP-RFT	33.59 $\pm$ 0.26	33.43	33.89	33.43
KTO	33.38 $\pm$ 0.98	34.19	33.66	32.30
DPO	33.18 $\pm$ 0.31	33.36	33.36	32.83
SimPO	23.55 $\pm$ 5.19	26.00	17.59	27.07

Table 19: Multi-seed results at 1.5B on GSM8K (3 seeds each). Rankings preserved across seeds.

Method	Original	Mean	σ	s42	s123	s456
SP-RFT	54.66	54.36	0.59	53.90	55.19	53.98
IPO	52.39	52.24	0.22	52.54	52.16	52.01
KTO	51.71	51.15	1.77	51.33	53.22	48.90
DPO	51.18	49.08	0.61	49.58	49.43	48.22
SimPO	43.44	38.67	1.78	36.85	38.06	41.09

- **DPO vs. SP-RFT:** $\Delta = +6.01$ pp, $p < 0.001$. **Significant.**
 - **SimPO vs. SP-RFT:** $\Delta = +5.94$ pp, $p < 0.001$. **Significant.**
 - **DPO vs. KTO:** $\Delta = +3.15$ pp, $p < 0.001$. **Significant.**
 - **DPO vs. IPO:** $\Delta = +2.99$ pp, $p = 0.029$. Not significant after correction.
 - **SP-RFT vs. IPO:** $\Delta = -3.01$ pp, $p = 0.018$. Not significant after correction.
 - **SP-RFT vs. KTO:** $\Delta = -2.86$ pp, $p = 0.006$. Not significant after correction (marginal).
 - **IPO vs. KTO:** $\Delta = +0.15$ pp, $p = 0.837$. Not significant.
- Three groups emerge: (1) DPO/SimPO cluster ($\sim 83.4\%$), (2) IPO/KTO cluster ($\sim 80.3\%$), (3) SP-RFT ($\sim 77.4\%$). The DPO/SimPO $>$ SP-RFT separation is robust ($p < 0.001$).

L Sampling-Based Evaluation

We re-evaluate 1.5B and 7B checkpoints using nucleus sampling (top- $p = 0.9$, temperature = 0.8) with 8 samples per problem, reporting majority-vote accuracy. Core rankings are preserved: at 7B, DPO and SimPO remain at the top, SP-RFT at the bottom. The inter-algorithm spread compresses slightly, consistent with sampling partially compensating for format compliance differences.

M Learning Rate Sensitivity at 7B

Three observations emerge. (1) The default 10^{-6} is optimal for both SP-RFT and DPO; no higher LR improves either algorithm, ruling out under-training as an explanation for SP-RFT’s 7B performance. (2) DPO is *more* LR-sensitive than SP-RFT: at 10^{-5} , DPO collapses to 16.15% (one seed hits 1.2%) while SP-RFT retains 53.12%. (3) The ranking inversion (DPO $>$ SP-RFT) holds at the independently optimal LR for each algorithm (10^{-6} for both), confirming it is not an artifact of hyperparameter tuning.

Note on the lower boundary. Our grid does not extend below 10^{-6} (e.g., to 5×10^{-7}). Because performance degrades *monotonically and steeply* at every LR above 10^{-6} —with DPO losing 27 pp at just $5 \times$ higher—the gradient strongly favors the conservative end, and 10^{-6} already matches the recommended LoRA LR for both Qwen 2.5 and Gemma 3 in their official fine-tuning guides. Nonetheless, we cannot rule out a marginal improvement at 5×10^{-7} ; if it exists, it would affect SP-RFT and DPO similarly and would not change the ranking inversion.

Table 20: Multi-seed results at 7B (LoRA) on GSM8K. Core algorithms have $N=5$ seeds; IPO/KTO have $N=3$.

Method	N	Mean	σ	s42	s456	s789	s1024	s1337
DPO	5	83.38	0.56	83.85	82.79	83.02	84.23	83.02
SimPO	5	83.32	1.79	85.82	81.50	82.64	85.06	81.58
IPO	3	80.39	0.92	81.35	—	80.67	—	79.15
KTO	3	80.24	0.16	80.21	—	80.44	—	80.06
SP-RFT	5	77.38	1.11	76.42	79.30	76.65	76.57	77.94

Table 21: Learning rate sensitivity for SP-RFT and DPO at 7B (LoRA) on GSM8K (%). Each cell reports mean $\pm$ std over 3 seeds (SP-RFT) or 2 seeds (DPO). The default LR = 10^{-6} is optimal for both algorithms; higher rates degrade performance monotonically.

LR	SP-RFT Mean	SP-RFT Seeds	DPO Mean	DPO Seeds
10^{-6} (default)	77.38 ± 1.11	5 seeds	83.38 ± 0.56	5 seeds
5×10^{-6}	66.57 ± 5.14	66.3 / 60.4 / 73.0	56.75 ± 8.68	65.4 / 48.1
10^{-5}	53.12 ± 4.90	59.3 / 47.3 / 52.8	16.15 ± 14.94	31.1 / 1.2
5×10^{-5}	30.20 ± 7.73	34.1 / 37.1 / 19.4	—	—
2×10^{-4}	0.23	1 seed	—	—

N Learning Rate Sensitivity at 1.5B

To directly address whether the 1.5B SP-RFT–DPO ranking is an artifact of the shared LR (10^{-6}), we sweep LR in $\{5 \times 10^{-7}, 10^{-6}, 5 \times 10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 1.5B with 3 seeds each (24 runs total).

Table 22: Learning rate sensitivity for SP-RFT and DPO at 1.5B (full FT) on GSM8K (%). Each cell reports mean $\pm$ population std over 3 seeds. SP-RFT leads at *every* LR. DPO collapses catastrophically at $\geq 5 \times 10^{-6}$. At independently optimal LRs, SP-RFT@ 10^{-5} (56.61%) beats DPO@ 5×10^{-7} (52.26%) by +4.35 pp.

LR	SP-RFT Mean $\pm\sigma$	SP-RFT Seeds	DPO Mean $\pm\sigma$	DPO Seeds
5×10^{-7}	53.83 ± 0.21	54.13 / 53.68 / 53.68	52.26 ± 0.52	52.46 / 52.77 / 51.55
10^{-6} (default)	54.36 ± 0.59	55.19 / 53.90 / 53.98	49.08 ± 0.61	49.43 / 49.58 / 48.22
5×10^{-6}	54.74 ± 1.21	56.25 / 53.30 / 54.66	16.00 ± 11.69	20.39 / 27.60 / 0.00 [†]
10^{-5}	56.61 ± 0.48	55.95 / 56.79 / 57.09	19.66 ± 1.81	20.62 / 21.23 / 17.13

[†]Complete collapse: model generates only degenerate output.

Four observations: **(1)** SP-RFT wins at every LR tested. **(2)** At independently optimal LRs, the gap widens to +4.35 pp. **(3)** DPO is dramatically more LR-sensitive at 1.5B than at 7B: the usable LR range for DPO shrinks from roughly one order of magnitude at 7B to half an order at 1.5B. **(4)** The 1.5B and 7B sweeps tell opposite stories about optimal LR, confirming that LR sensitivity itself is scale-dependent.

O Learning Rate Sensitivity at 3B

The 3B collapse (Section 4.1) is fully recoverable with a LR sweep. We sweep LR in $\{5 \times 10^{-7}, 5 \times 10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 3B (full FT) with 3 seeds each (18 runs).

Three observations: **(1)** The 3B collapse is fully recoverable for SP-RFT: at 10^{-5} , it reaches 55.70%—comparable to 1.5B Instruct (54.36%). **(2)** SP-RFT wins at every LR, preserving the small-scale ranking. **(3)** DPO is dramatically more LR-sensitive than SP-RFT at 3B, mirroring the pattern at 1.5B and 7B.

Table 23: Learning rate sensitivity for SP-RFT and DPO at 3B (full FT) on GSM8K (%). Each cell reports mean $\pm$ population std over 3 seeds. SP-RFT recovers to 55.7% at 10^{-5} —comparable to 1.5B Instruct—confirming the default-LR collapse was a hyperparameter pathology. SP-RFT leads at every LR.

LR	SP-RFT Mean $\pm\sigma$	SP-RFT Seeds	DPO Mean $\pm\sigma$	DPO Seeds
5×10^{-7}	17.79 ± 0.16	17.82 / 17.97 / 17.59	15.77 ± 0.91	14.56 / 16.00 / 16.76
10^{-6} (default)	18.35^b	—	14.44 ± 2.01	16.45 / 14.44 / 12.42
5×10^{-6}	41.80 ± 3.84	40.26 / 38.06 / 47.08	34.55 ± 4.52	33.81 / 29.42 / 40.41
10^{-5}	55.70 ± 1.09	54.21 / 56.79 / 56.10	16.30 ± 7.34	5.91 / 21.46 / 21.53

^bSingle seed from Table 1.

515 P KL Regularization Ablation for SGRPO

Table 24: SGRPO at 7B with and without KL regularization on GSM8K (%).

Method	λ_{KL}	GSM8K (%)	Δ vs. DPO
SGRPO (default)	0	80.59	−2.79
SGRPO (regularized)	0.01	80.74	−2.64
DPO (reference)	$\beta = 0.1$	83.38 ± 0.56	—

516 Adding KL regularization yields only +0.15 pp, confirming the online–offline gap is genuine rather
517 than a regularization artifact.

518 Q Base vs. Instruct Initialization at 7B

Table 25: Base vs. Instruct initialization at 7B on GSM8K (%). The spread collapses from 6.00 pp to 1.82 pp. SP-RFT leads from Base, reversing the Instruct ranking.

Algorithm	From Base (7B)	From Instruct (7B)	Δ
SP-RFT	82.49	77.38 ± 1.11	+5.11
DPO	80.67	83.38 ± 0.56	−2.71
SimPO	80.82	83.32 ± 1.79	−2.50
<i>Spread</i>	<i>1.82 pp</i>	<i>6.00 pp</i>	

519 From Base initialization at 7B, the ranking (SP-RFT > SimPO > DPO) differs from both the 1.5B
520 Base ranking (SimPO > DPO > SP-RFT) and the 7B Instruct ranking (DPO > SimPO $\gg$ SP-RFT).
521 A residual ranking shift between scales persists even from Base, but at ≤ 1.82 pp—far smaller than
522 the 6.00 pp Instruct spread. The key finding is that initialization accounts for the majority of inter-
523 algorithm variance ($3.3\times$ compression at 7B, $14.7\times$ at 1.5B), even though a small scale-dependent
524 residual remains.

525 R Hyperparameter Sensitivity

526 S 3B Flexible Extraction Analysis

527 Under flexible extraction, the 3B base model achieves 62.93%—comparable to the 1.5B Instruct
528 model’s strict-match performance. This reveals that the Qwen 2.5 3B Instruct model can *solve* GSM8K
529 problems at a competitive rate but fails to produce answers in the expected format. Post-training
530 partially addresses this: SP-RFT reduces the format gap from −52.2 pp (base) to −26.2 pp.

Table 26: β sensitivity at 1.5B on GSM8K (%). Both variants are stable across a $50\times$ range; default $\beta = 0.1$ is near-optimal.

Variant	$\beta = 0.01$	$\beta = 0.1$	$\beta = 0.5$	Range
DPOP	54.66	55.19	53.30	1.89 pp
ORPO	54.81	54.44	53.83	0.98 pp
<i>Vanilla DPO</i>	<i>49.76$\pm$2.27 (variant sweep)</i>			

T General Benchmark Results

Table 27 reports ARC-Challenge, HellaSwag, and WinoGrande accuracy across scales and architectures. At every scale and architecture, the spread is <2 pp—confirming that post-training algorithms do not meaningfully differ on these benchmarks. **This is a critical scope qualifier for our main findings:** the ranking inversion documented in this paper is specific to mathematical reasoning (GSM8K, MATH). On these general benchmarks, no inversion is observable, and algorithm choice is essentially irrelevant.

Caveat: benchmark format. ARC-C, HellaSwag, and WinoGrande are multiple-choice or perplexity-based evaluations, not open-ended generative tasks. Since our analysis identifies format compliance—a generative property—as the key differentiator at small scales (Section 4.4), the absence of algorithmic differences on non-generative benchmarks is expected and does not rule out the possibility that inversions occur on other *generative* tasks. We leave this to future work.

Table 27: General benchmark accuracy (%) after post-training. Spread remains <2 pp across all scales and architectures, confirming that algorithmic differences are task-specific.

Architecture	Algorithm	ARC-C	HellaSwag	WinoGrande	Avg
Gemma 1B	Base	38.7	57.7	58.9	51.8
	SP-RFT	39.1	57.9	59.1	52.0
	DPO	38.7	57.6	58.8	51.7
	SimPO	39.6	57.2	57.6	51.5
Gemma 12B	Base	61.1	81.9	74.6	72.5
	SP-RFT	60.1	81.7	74.5	72.1
	DPO	58.4	81.2	74.6	71.4
	SimPO	56.7	81.3	73.6	70.5
Qwen2.5-7B FT	Base	55.2	80.5	71.1	68.9
	SP-RFT	56.1	80.6	71.0	69.2
	DPO	55.0	80.6	70.4	68.7
	SimPO	54.1	80.6	69.9	68.2
Qwen2.5-14B	Base	62.4	84.4	76.0	74.3
	SP-RFT	61.9	84.5	75.2	73.9
	DPO	61.8	84.4	75.4	73.9
	SimPO	61.7	84.5	74.8	73.7

NeurIPS Paper Checklist

1. Claims

Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction make three claims: (1) rankings invert with scale, (2) initialization dominates algorithm choice, and (3) apparent differences have two regimes. Each is supported by a dedicated results subsection (Sec. 4.2, Sec. 4.1, Sec. 4.3, and Sec. 4.4), with formal statistical testing ($p < 0.001$, Bonferroni-corrected) and cross-architecture replication (Gemma 3). The scope is explicitly limited to mathematical reasoning on two model families.

2. Limitations

Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: A dedicated **Limitations** paragraph appears in the Discussion (page 9 approximately): two model families, math-only generative tasks, MC-only general benchmarks, asymmetric fixed-dataset control, and noise-floor (0.5B) / LR-pathology (3B) caveats. Appendices K and O discuss these in detail.

3. Theory assumptions and proofs

For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is an empirical study of post-training algorithms; it presents no formal theorems requiring proofs. Statistical procedures (Welch’s t -test, Bonferroni correction) are standard and cited.

4. Experimental result reproducibility

Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All training configurations (learning rates, batch sizes, LoRA rank, KL coefficients, optimizer, GPUs/run) are listed in Appendix B (Table in Sec. “Training Configurations”). Datasets (GSM8K, MATH, ARC-C, HellaSwag, WinoGrande), models (Qwen 2.5 0.5B–14B, Gemma 3 1B–12B), seed counts, and evaluation protocol are specified. The repository (oxRL) is referenced; code and data will be released upon acceptance per the Limitations statement.

5. Open access to data and code

Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All datasets used (GSM8K, MATH, ARC-C, HellaSwag, WinoGrande) are public benchmarks with standard splits. The oxRL training framework is open-source; the repository URL and pre-trained checkpoints will be deanonymized upon acceptance. Supplementary material includes training configs and evaluation scripts. Self-play training data (generated rollouts) will be released alongside checkpoints to enable exact replication.

6. Experimental setting/details

Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 (“Methods and Experimental Setup”) specifies the protocol; Appendix B lists per-algorithm hyperparameters and learning rates. Learning rate sensitivity is analyzed at 1.5B, 3B, and 7B (Appendices O and the LR-sweep appendices). Test protocols (strict-match, sampling-based) are documented in Appendix L.

595 **7. Experiment statistical significance**

596 Does the paper report error bars suitably and correctly defined or other appropriate informa-
597 tion about the statistical significance of the experiments?

598 **Answer:** [Yes]

599 **Justification:** Up to $N = 5$ seeds are run for headline comparisons (1.5B and 7B SP-
600 RFT/DPO/SimPO); 3 seeds for cross-architecture (Gemma 3) and 14B runs. Standard
601 deviations are reported throughout (e.g., $83.38\% \pm 0.56$). Welch’s t -tests with Bonferroni
602 correction are applied; the $\alpha = 0.005$ family-wise threshold and 80% power calculation are
603 stated explicitly (Appendix K, DPO variant table caption).

604 **8. Experiments compute resources**

605 For each experiment, does the paper provide sufficient information on the computer resources
606 (type of compute workers, memory, time of execution) needed to reproduce the experiments?

607 **Answer:** [Yes]

608 **Justification:** Appendix I (“Compute Budget”) gives a per-category breakdown across
609 ~ 350 training runs: GPU type (H100 80GB), GPUs/run, time/run, and total GPU-hours per
610 category. Grand total: ~ 940 GPU-hours.

611 **9. Code of ethics**

612 Does the research conducted in the paper conform, in every respect, with the NeurIPS Code
613 of Ethics?

614 **Answer:** [Yes]

615 **Justification:** The work uses publicly released model weights (Qwen 2.5, Gemma 3) and
616 public benchmark datasets (GSM8K, MATH, ARC, HellaSwag, WinoGrande) under their
617 respective licenses. No human subjects, personal data, or sensitive deployments are involved.
618 The study is a methodological audit of post-training algorithms.

619 **10. Broader impacts**

620 Does the paper discuss both potential positive societal impacts and negative societal impacts
621 of the work performed?

622 **Answer:** [Yes]

623 **Justification:** The Discussion notes that the central finding—small-scale algorithm rankings
624 invert at deployment scale—has direct positive impact for the post-training research com-
625 munity by exposing a methodological hazard and arguing for deployment-scale validation.
626 Negative impacts are minimal: the paper releases no new generative capability beyond what
627 is already publicly available; it studies algorithm-selection methodology, not new model
628 artifacts.

629 **11. Safeguards**

630 Does the paper describe safeguards that have been put in place for responsible release of data
631 or models with a high risk for misuse (e.g., pretrained language models, image generators,
632 or scraped datasets)?

633 **Answer:** [N/A]

634 **Justification:** The paper releases fine-tuned math-reasoning checkpoints derived from
635 already-public base models (Qwen 2.5, Gemma 3). Math-task fine-tunes do not introduce
636 new misuse risks beyond those of the base models, which already shipped with their own
637 safety releases.

638 **12. Licenses for existing assets**

639 Are the creators or original owners of assets (e.g., code, data, models) used in the paper
640 properly credited and are the license and terms of use explicitly mentioned and respected?

641 **Answer:** [Yes]

642 **Justification:** All base models (Qwen 2.5 (Qwen Team, 2025), Gemma 3 (Gemma Team,
643 2025)) and benchmarks (GSM8K, MATH, ARC, HellaSwag, WinoGrande) are cited.
644 Qwen 2.5 is released under Apache 2.0 (0.5B/1.5B/7B/14B) and Qwen-Research license
645 (3B); Gemma 3 is under the Gemma Terms of Use; the benchmarks are MIT/Apache 2.0/CC-
646 BY. All uses are within license terms (research, non-commercial evaluation).

- 647 **13. New assets**
648 Are new assets introduced in the paper well documented and is the documentation provided
649 alongside the assets?
650 **Answer:** [Yes]
651 **Justification:** The oxRL framework (release planned upon deanonymization) ships with
652 a README documenting training, evaluation, and reproduction commands; a Croissant
653 metadata file documents released self-play datasets including biases, collection, and PII
654 fields. Per-checkpoint model cards specify training algorithm, base model, hyperparameters,
655 and intended use.
- 656 **14. Crowdsourcing and research with human subjects**
657 For crowdsourcing experiments and research with human subjects, does the paper include
658 the full text of instructions given to participants and screenshots, if applicable, as well as
659 details about compensation (if any)?
660 **Answer:** [N/A]
661 **Justification:** No crowdsourcing or human-subject experiments. All evaluation uses auto-
662 mated metric extraction on public benchmarks.
- 663 **15. Institutional review board (IRB) approvals or equivalent for research with human**
664 **subjects**
665 Does the paper describe potential risks incurred by study participants, whether such risks
666 were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals
667 (or an equivalent approval/review based on the requirements of your country or institution)
668 were obtained?
669 **Answer:** [N/A]
670 **Justification:** No human-subject research; IRB approval not applicable.
- 671 **16. Declaration of LLM usage**
672 Does the paper describe the usage of LLMs if it is an important, original, or non-standard
673 component of the core methods in this research?
674 **Answer:** [Yes]
675 **Justification:** LLMs are the object of study: the paper post-trains and evaluates open-weight
676 LLMs (Qwen 2.5 0.5B–14B, Gemma 3 1B–12B). Their roles—as base policies, as self-play
677 trajectory generators (SP-RFT), and as evaluation targets—are documented in Sec. 3 and
678 Appendix B. LLMs were not used as part of the writing methodology beyond standard
679 editing assistance.

680 **U Discussion-Period Additions**

681 The following experiments were run during the NeurIPS discussion period in response to reviewer
682 requests. Raw evaluation JSONs accompany the released artifacts.

683 **U.1 Qwen2.5-32B Deployment Scale**

684 Identical protocol to the 14B deployment runs: self-play data generated by the 32B model itself,
685 shared hyperparameters, GSM8K 8-shot strict-match, greedy. Of 7,473 prompts, only **816** yield
686 informative preference pairs at 32B (6,657 discarded because all 16 samples agree), versus 1,332 at
687 14B — a direct measurement of the RFT ceiling closing.

688 **U.2 Fixed-Dataset Control on Gemma 3**

689 Gemma-3-1B trained on Gemma-3-12B-generated self-play data (3 seeds, identical hyperparameters);
690 own-data checkpoints re-evaluated under the identical protocol (GSM8K 8-shot strict-match, vLLM
691 backend). Unlike Qwen — where 14B-quality data preserves the small-scale SP-RFT>DPO ordering
692 (Section 4.5) — strong-teacher data *flips* the ordering at Gemma-1B, consistent with a capability-gap
693 effect: SP-RFT imitates trajectories the 1B model cannot reproduce, while DPO’s contrastive signal

Table 28: Qwen2.5-32B on GSM8K (8-shot strict-match; mean $\pm\sigma$ over 3 seeds). The small-scale ranking is fully reversed: SimPO (worst at $\leq 1.5\text{B}$) is best; SP-RFT (best at $\leq 3\text{B}$) gains $+0.30 \pm 0.23$ pp over base — statistically zero. SP-RFT’s gain over base collapses $+23.2$ pp (1.5B) $\rightarrow +1.6$ (7B) $\rightarrow +0.3$ (32B); DPO $>$ SP-RFT at $p < 0.01$ (Welch).

Method	GSM8K (%)	Δ vs. Base
Base (Instruct)	80.52	—
SP-RFT	$80.82_{\pm 0.23}$	+0.30
DPO	$82.23_{\pm 0.04}$	+1.71
SimPO	$83.42_{\pm 1.14}$	+2.90

694 degrades gracefully. The fixed-dataset conclusion of Section 4.5 is therefore scoped to Qwen; across
695 families, rankings are unstable across scale, architecture, *and data provenance*.

Table 29: Gemma-3-1B fixed-dataset control (GSM8K 8-shot strict-match, 3 seeds).

Training data	SP-RFT	DPO	Winner
Own (1B) self-play	$26.91_{\pm 0.55}$	$22.82_{\pm 2.08}$	SP-RFT (+4.1 pp)
12B-generated	$21.41_{\pm 1.01}$	$24.69_{\pm 1.01}$	DPO (+3.3 pp)

696 U.3 Previously-Missing 3B Cells

697 IPO, KTO, and SimPO at 3B under the shared default LR (10^{-6}), 3 seeds each (GSM8K 8-shot
698 strict-match): IPO $24.92_{\pm 0.85}$, KTO $23.68_{\pm 0.87}$, SimPO $9.60_{\pm 3.84}$. All methods remain far below the
699 base checkpoint’s 62.9% flexible-extract ability, confirming the 3B default-LR pathology (Section 4.1)
700 is checkpoint-wide rather than method-specific, with SimPO again the most fragile.